

Quantum Error Correction

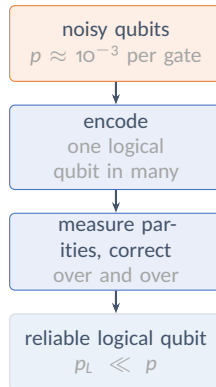
Why we need it, and the idea that makes it possible

Lecture 10

Term 261

Agenda

- 1 Why errors matter: the numbers
- 2 Classical error correction in one picture
- 3 Why the quantum case looks impossible
- 4 The idea that makes it possible
- 5 The QEC cycle, and a first example
- 6 The threshold, and its price
- 7 Thirty years of milestones; the road to L16



Why errors matter: the numbers

Best two-qubit gates (as of 2026)

Device	error / gate
Google Willow, QEC chip (CZ)	3.3×10^{-3}
Quantinuum Helios (98 trapped ions)	7.9×10^{-4}
IonQ / Oxford Ionics (2 ions)	8.4×10^{-5}

Rule of thumb for a large machine: $p \approx 10^{-3}$.

Factoring RSA-2048 (Gidney 2025)

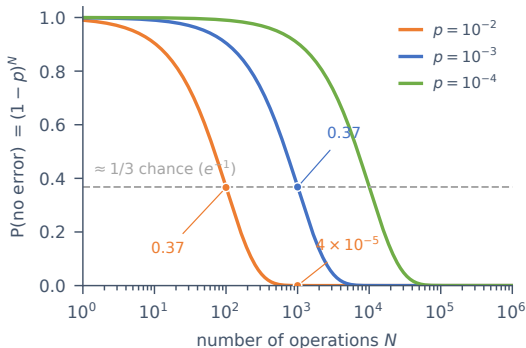
- 1399 logical qubits, 6.5×10^9 Toffoli gates
- $< 10^6$ noisy qubits at $p = 10^{-3}$, < 1 week (2021 estimate: 2×10^7 qubits)
- needs logical error $\sim 10^{-15}$ per qubit per round



The catch: run those 6.5×10^9 gates unprotected at $p = 10^{-3}$ and you expect 6.5×10^6 errors. No engineering closes a gap of 10^{12} ; error correction must.

Google Quantum AI, Willow spec sheet (2024). Ransford et al., arXiv:2511.05465 (2025). Hughes et al., arXiv:2510.17286 (2025). Gidney, arXiv:2505.15917 (2025). Gidney & Ekerå, Quantum 5, 433 (2021).

The arithmetic of failure



If each operation fails independently with probability p ,

$$P(\text{no error in } N \text{ ops}) = (1 - p)^N \approx e^{-pN}.$$

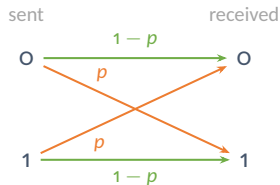
p	N	$(1 - p)^N$
1%	100	$0.99^{100} = 0.37$
1%	1000	$0.99^{1000} \approx 4 \times 10^{-5}$
0.1%	1000	$0.999^{1000} = 0.37$

To get a one-in-three chance back at 1000 operations, p must improve **tenfold**.

Takeaway: A computer with error rate p can run about $1/p$ operations. Useful algorithms need billions, so we need $p_L \sim 10^{-10}$ to 10^{-15} .

Classical error correction in one picture

The bit-flip channel

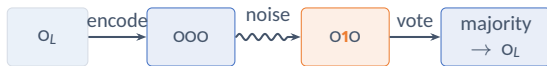


p = bit error rate (BER).

At $p = 10^{-2}$: send 1000 bits, about **10 arrive wrong**.

Two ideas to keep: **redundancy** (logical bit $\rightarrow$ several physical bits) and $p_L \propto p^2$ (errors must conspire).
L12 does this code in full.

The 3-bit repetition code



The vote fails only if **two or three** bits flip:

$$p_L = 3p^2(1-p) + p^3 = 3p^2 - 2p^3$$

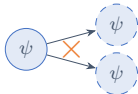
At $p = 0.01$

$$p_L = 3(10^{-4}) - 2(10^{-6}) = 2.98 \times 10^{-4} \text{ — about } 34\times \text{ better.}$$

Why the quantum case looks impossible

Copy the qubit three times and take a vote? Three obstacles stand in the way.

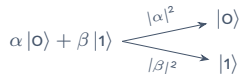
1. No-cloning



No U with $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$.

Proof. If it copies $|\psi\rangle$ and $|\phi\rangle$, take inner products: $\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$, so $\langle\psi|\phi\rangle \in \{0, 1\}$. $\square$

2. Measurement destroys



A majority vote must *read* the bits. Reading a qubit collapses it, and α, β — the very thing we protect — are gone. One measurement reveals one bit, never the amplitudes.

3. Errors are continuous



A bit flips or it does not. A qubit can be rotated by *any* small angle θ : a continuum of errors.

And **phase flips** Z :

$\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|0\rangle - \beta|1\rangle$, which have no classical analogue.

■ In 1995 many believed these were fatal. Each one is dissolved, not solved — the next three slides.

Move 1: redundancy by entanglement, not by copying

Forbidden: copy

$$\alpha |0\rangle + \beta |1\rangle \not\mapsto (\alpha |0\rangle + \beta |1\rangle)^{\otimes 3}$$

Expanding gives 8 terms, with amplitudes $\alpha^3, \alpha^2\beta, \dots$ — not linear in (α, β) , so no unitary does it.

Allowed: encode

$$\alpha |0\rangle + \beta |1\rangle \mapsto \alpha |000\rangle + \beta |111\rangle = \alpha |0_L\rangle + \beta |1_L\rangle$$

Linear in (α, β) : two CNOTs do it (L12).

Takeaway: Redundancy without copying: encode one logical qubit into an entangled state of many physical qubits. Vocabulary: **logical** qubit $|\psi_L\rangle$, **physical** (data) qubits.

one logical qubit, spread over three



each qubit alone: $\rho_j = \begin{pmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{pmatrix}$

The information lives in the correlations

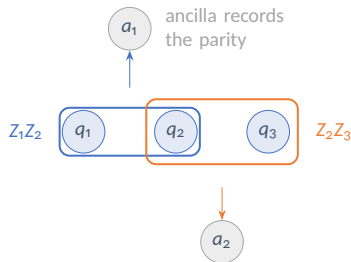
No single qubit holds a copy of $|\psi\rangle$: the relative phase of α, β is invisible to each qubit alone.

Move 2: measure parities, never the data

Ask a *collective* question: “do qubits 1 and 2 agree?” That is the observable Z_1Z_2 .

$$Z_1Z_2(\alpha|000\rangle + \beta|111\rangle) = +(\alpha|000\rangle + \beta|111\rangle)$$

- Both branches have the same parity, so the outcome is **+1 with certainty**
- A certain outcome carries no information about α, β — and does not disturb them
- After an error the outcome changes: it reports the **error**, not the data



Syndrome

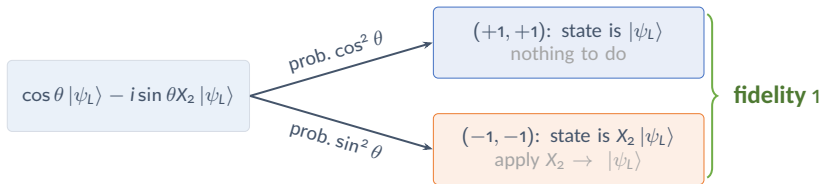
The list of parity outcomes, e.g. $(Z_1Z_2, Z_2Z_3) = (+1, +1)$, is the **syndrome**. It is read out through extra **ancilla** qubits, so the data qubits are never measured.

Move 3: measuring the syndrome makes errors discrete

A small, *continuous* over-rotation of qubit 2 by θ :

$$e^{-i\theta X_2} |\psi_L\rangle = \cos \theta |\psi_L\rangle - i \sin \theta X_2 |\psi_L\rangle$$

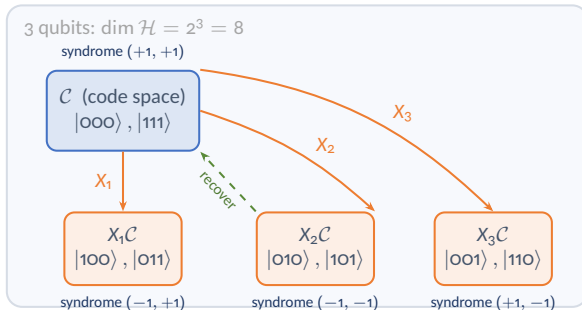
Now measure the syndrome. The two terms lie in different syndrome sectors, so the state is forced into one of them:



Worked, $\theta = 0.1$: outcome $(+1, +1)$ with probability $\cos^2 0.1 = 0.990$, outcome $(-1, -1)$ with probability $\sin^2 0.1 = 0.010$. Both end in $|\psi_L\rangle$ exactly.

Takeaway: The measurement turns a continuum of errors into a discrete choice: “no error” or “ X_2 ”. Correct a finite set of Pauli errors and you correct every error (L12, L13 prove it).

The geometric picture: code space and error spaces

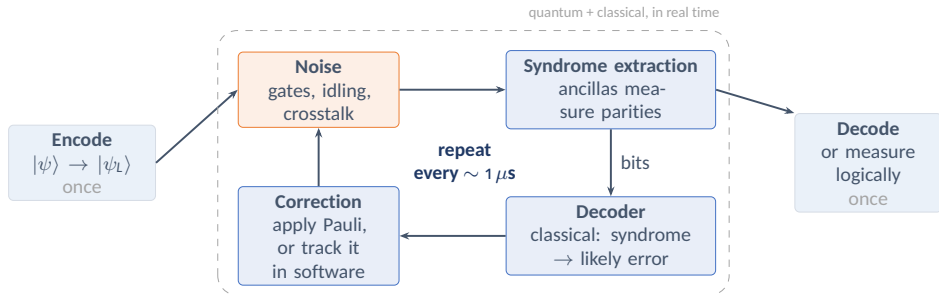


- The code is a **subspace** $\mathcal{C}$, not a set of copies
- Each correctable error moves $\mathcal{C}$ to an **orthogonal** subspace
- The syndrome asks only “which subspace?” — it never looks inside
- Recovery maps back; α, β ride along untouched

Count: 4 subspaces $\times \dim 2 = 8$. The three single flips fill $\mathcal{H}$ exactly.

Takeaway: Quantum error correction = choose a subspace whose error images are mutually orthogonal. L13 turns this into the Knill-Laflamme conditions.

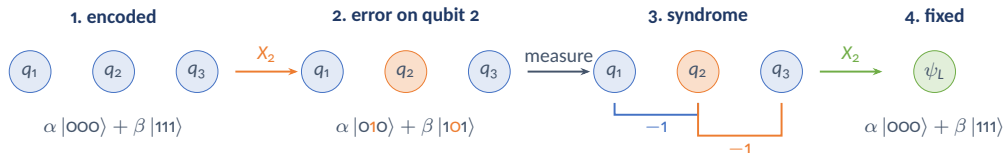
The QEC cycle



Vocabulary

- **logical** qubit: the protected one, $|\psi_L\rangle$
- **physical** (data) qubits: the hardware carrying it
- **ancilla** (measure) qubits: read out parities
- **syndrome**: the list of parity outcomes
- **decoder**: classical algorithm, syndrome → correction
- **cycle** (round): one pass of the loop

A first example, in pictures: the 3-qubit bit-flip code

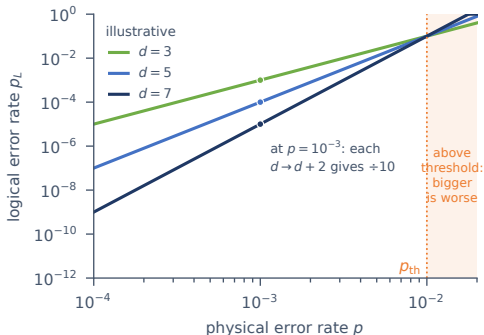


$(Z_1 Z_2, Z_2 Z_3)$	diagnosis
$(+1, +1)$	no error
$(-1, +1)$	X_1
$(-1, -1)$	X_2
$(+1, -1)$	X_3

- $|010\rangle$ and $|101\rangle$ give the **same** parities: $(-1, -1)$
- So the outcome is certain; the superposition survives
- We learned *where* the error is, nothing about α, β

The catch: this code is blind to phase flips: Z_1 gives $\alpha |000\rangle - \beta |111\rangle$ with syndrome $(+1, +1)$. L12 fixes that, ending with Shor's 9-qubit code.

The threshold: bigger codes help, if $p < p_{\text{th}}$



illustrative values $A = 0.1$, $p_{\text{th}} = 10^{-2}$

A code of **distance** d corrects up to $\lfloor (d-1)/2 \rfloor$ errors; failure needs about $(d+1)/2$ of them. So, roughly,

$$p_L \approx A \left(\frac{p}{p_{\text{th}}} \right)^{(d+1)/2}.$$

- $p < p_{\text{th}}$: each step $d \rightarrow d+2$ divides p_L by $\Lambda = p_{\text{th}}/p$
- $p > p_{\text{th}}$: more qubits, more ways to fail — bigger is worse

Worked at $p = 10^{-3}$ ($\Lambda = 10$):

$d = 3, 5, 7 \Rightarrow p_L = 10^{-3}, 10^{-4}, 10^{-5}$;

$d = 27$ reaches 10^{-15} .

Takeaway: Below threshold, logical errors fall *exponentially* in d , while the qubit count grows only polynomially. That is why QEC can close a 10^{12} gap.

The threshold theorem — and where experiments are

Threshold theorem (informal)

If every physical gate, preparation and measurement fails independently with probability $p < p_{\text{th}}$, any ideal circuit of N gates can be run with final error $\leq \varepsilon$ using only $\mathcal{O}(N \text{polylog}(N/\varepsilon))$ physical gates.

Proved independently, 1996–1998

- Aharonov & Ben-Or (STOC 1997)
- Knill, Laflamme & Zurek (Science 1998)
- Kitaev (Russ. Math. Surveys 1997)

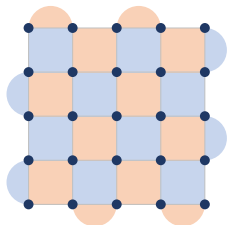
Early thresholds were tiny ($\sim 10^{-6}$); the surface code tolerates about 1% (Fowler et al. 2012).

Aharonov & Ben-Or, STOC'97 pp. 176–188; Knill, Laflamme, Zurek, Science 279, 342 (1998); Kitaev, Russ. Math. Surv. 52, 1191 (1997); Google Quantum AI, Nature 638, 920 (2025).

Below threshold, measured (Google Willow)

- Surface-code memories up to $d = 7$ (101 qubits)
- $\Lambda = 2.14 \pm 0.02$: each $d \rightarrow d + 2$ divides p_L by 2.14
- $d = 7$: 0.143% error per cycle, outliving the best physical qubit by $2.4\times$
- Real-time decoding at $d = 5$; $1.1 \mu\text{s}$ cycle

The price: many physical qubits per logical qubit



$d = 5$: ● 25 data qubits;
24 checks ■ X-type, ■ Z-type,
each with its own measure qubit

Surface code (L16): d^2 data + $(d^2 - 1)$ measure qubits $\approx 2d^2$.

d	data	measure	total per logical qubit
3	9	8	17
5	25	24	49
7	49	48	97
25	625	624	1249

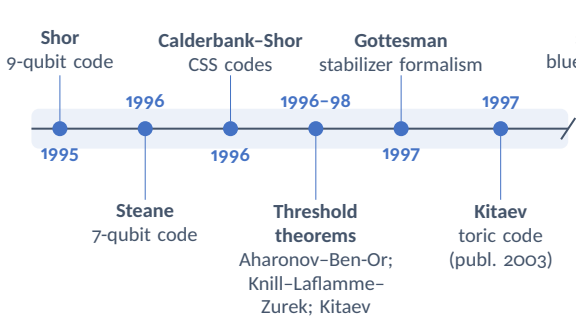
RSA-2048 needs $d = 25$: 1399 logical qubits $\times \sim 1.3\text{k} \approx 1.9$ million. Gidney gets under 10^6 with denser idle storage (yoked codes) and cheaper magic states.

qLDPC codes cut this: IBM's gross code $[[144, 12, 12]]$ keeps 12 logical qubits in 288 physical (24 each), vs. ~ 3000 for the surface code.

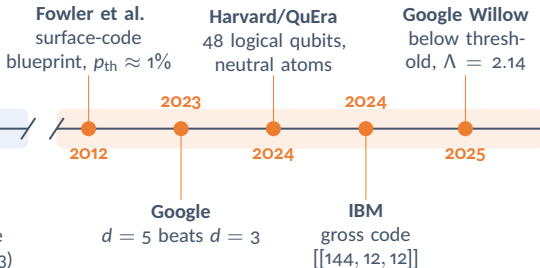
Fowler et al., PRA 86, 032324 (2012); Gidney, arXiv:2505.15917 (2025); Bravyi et al., Nature 627, 778 (2024). Gidney budgets $2(d+1)^2 = 1352$ per patch incl. routing.

Thirty years of quantum error correction

theory: the idea is possible



experiment: the idea works



Shor, PRA 52, R2493 (1995); Steane, PRL 77, 793 (1996); Calderbank & Shor, PRA 54, 1098 (1996); Gottesman, PhD thesis, quant-ph/9705052 (1997); Kitaev, Ann. Phys. 303, 2 (2003); Fowler et al., PRA 86, 032324 (2012); Google Quantum AI, Nature 614, 676 (2023); Bluvstein et al., Nature 626, 58 (2024); Bravyi et al., Nature 627, 778 (2024); Google Quantum AI, Nature 638, 920 (2025).

The road ahead: L10–L16

Lecture	Topic	The question it answers
L10	Introduction	Why do we need QEC, and why is it possible at all?
L11	Noise models	What can go wrong with a qubit, and how do we write it down?
L12	Repetition → Shor	How do we correct bit flips, then phase flips, then both?
L13	QEC concepts	What makes a code good? $[[n, k, d]]$, Knill–Laflamme, bounds
L14	Stabilizer formalism	How do we describe a code by the Paulis that fix it?
L15	CSS codes	How do classical linear codes build quantum ones? (Steane)
L16	Surface → qLDPC	Which codes will run on hardware, and at what cost?

Today's three moves — **entangle, measure parities, let measurement discretize** — return in every one of them.

Summary

- Physical gates fail at $p \approx 10^{-3}$; useful algorithms need $p_L \sim 10^{-15}$. With $(1 - p)^N$, a machine runs only about $1/p$ operations
- Classically, repetition and a majority vote give $p_L = 3p^2 - 2p^3$ (2.98×10^{-4} at $p = 0.01$)
- Quantum obstacles: no-cloning, destructive measurement, continuous and phase errors
- The way out: encode into an entangled subspace, measure only parities (the syndrome), and let that measurement discretize the error
- The QEC cycle: encode $\rightarrow$ [noise $\rightarrow$ syndrome $\rightarrow$ decode $\rightarrow$ correct] * $\rightarrow$ read out
- Below threshold, $p_L \approx A(p/p_{\text{th}})^{(d+1)/2}$ falls exponentially in d — demonstrated in 2024/25 ($\Lambda = 2.14$), at a cost of $\sim 2d^2$ qubits per logical qubit

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Founding papers

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Threshold theorems

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